Project: Hunter Executive Summary & System Documentation

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Executive Summary

The "Hunter" project is an advanced, automated network forensics and threat detection tool developed for the Kali Linux environment. Its primary objective is to monitor live network traffic, identify connections to malicious indicators (IOCs), and analyze suspicious files transferred over the network.

Operating passively, the system dynamically pulls Threat Intelligence Feeds and integrates advanced tools such as Tshark for packet analysis and the VirusTotal API for file signature verification. This tool provides a comprehensive solution for Incident Response (SOC/IR) teams through real-time alerts and accurate, forensically sound event logging.

System Functions Documentation

1. Network Analysis This module is responsible for monitoring live network traffic. It utilizes Tshark to extract real-time data such as IP addresses (source and destination), DNS requests, and HTTP hosts. Simultaneously, the function dynamically downloads updated blacklists (Indicators of Compromise) from verified global sources and compares the live traffic against them. When a connection to a malicious IP or domain is detected, the system triggers an immediate alert.

2. File Analysis This function operates in the background, capturing traffic into rotating PCAP files. A dedicated background process reads these files, extracts transferred network objects (such as HTTP files), and filters files smaller than 1MB (per the project requirements). Every file that passes the filter undergoes SHA-256 hashing, and its signature is automatically sent to the VirusTotal API for scanning. Files identified as malicious are saved in an evidence directory for further investigation.

3. Log and Audit A critical component for investigation and forensics. Every anomalous detection (whether a malicious network connection or an infected file) is forwarded to the logging function. This function creates a standardized record containing a precise timestamp, source address, destination address (or file details), and the nature of the threat. The data is securely saved to a central log file, ensuring all events are documented and retained even after the software is closed.

4. Monitoring (Live Monitoring) The Command Line Interface (CLI) of the tool was specifically designed for cybersecurity personnel operating in the field. Upon execution, a clean screen displays the project details and the listening status ("//Hunter Live//"). The system provides continuous, color-coded visual alerts for detected vulnerabilities. Additionally, a "Heartbeat" mechanism is integrated, confirming to the operator that the system is running and scanning properly in the background without cluttering the main alert screen.